Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Vineet Palsule	vpalsule@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

I plan on making a study planner for college students to use for better organization and planning for everyday college students' lives.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

A popular example for a lot of college students is Google Calendar. It helps students plan out the upcoming days and having a calendar can function as a countdown until something or as a to-do list for someone that specific day.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I want this website to be useful for college students to organize their assignments and specific tasks. Users can enter their courses, assignments, due dates, time expected to take, and importance/weightage of the assignment to help plan out how college students can plan their assignments to fit the time constraints of the assignment.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The goal of my program is to take in user input about assignment information and process the data using lists, dataclasses, loops, if-statements, and drafter routes/pages. The output will contain an organized plan for a student to use to complete their assignments and plan their day according to the assignment's due dates and time requirements.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will provide the assignment name, course name, due date, time expected to complete, weightage of the assignments, and a checkbox for if the assignment is completed to yet to be completed.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program will take the assignments in a list and use loops and if-statements to take in specific details of those assignments. It will compare due dates, time expected to complete, and weightage of the assignment from the uncompleted assignments and filter them out to give a specific plan for the student to use to go about completing their assignments.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get a recommended schedule to help them plan out when a student should complete their assignments given the details of the assignments.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

The core “things” my program needs to track are “Assignments”, “Course”, and “Study Plan.” “Assignments” takes in the specific details of each assignment, “Course” associates the assignment to that specific course, and “Study Plan” takes in all the information from “Assignment” and “Course” to give the user a study plan.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need: Assignment name, course name, due date of assignment, amount of points/category of assignment, and expected time to complete the assignment.

Details I will ignore: Location of the class and information about the class teacher/professor

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The most difficult or technically complex part of this project will be making the study plan from the details of the assignment/course and deciding how the study plan should be built from it. Also using drafter could be difficult in making the website for the study planner.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

While I might use an LLM to help brainstorm ideas, I will try to only take suggestions from it and try to make it my own to avoid "shallow learning." I will also try to test my code in many different ways from the suggestions given instead of directly asking for solutions. I will try to distribute the time each week for the project evenly across each day to avoid myself rushing in the end and falling for the temptation at that point for using AI.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My ultimate goal is to build a study planner website that can help out college students like myself. While I hope to practice my coding skills more in this project, it is also a goal of mine to take those skills and solve a real-world problem.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

Prompt:

I want to make a study planner website for college students to use for better organization and efficiency for college students' lives. What are 3 common problems students face when going about managing and completing assignments, and what are 3 simple features to help solve these problems?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

AI suggested making a task breakdown, assignment dashboard, and daily study planner to be the core of the website. Websites like "MyStudyLife" and "Google Calendar" are examples of existing solutions to this topic.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

Can we simplify this project so it is easier for a first-year computer science student to make using basic Python concepts and drafter?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

My project has changed to become more focused on assignment organization and deadline prioritization instead of making a full schedule for a college student to use. I can make it where I give the program the assignment and deadline information and then the student can plan their work around it.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will try to refer to earlier course material and examples to help refresh earlier Python concepts. I can also try to ask TA's and professors for help if I am stuck on a part of my project.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

If I get an error from code given by LLM, I will try to read carefully and see where the error occurs and try to test my code multiple times to catch any errors to make my final product as error-free as possible.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously,

these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

QUESTION: I want to make a study planner website for college students to use for better organization and efficiency for college students' lives. What are 3 common problems students face when going about managing and completing assignments, and what are 3 simple features to help solve these problems?

AI: For your study planner website, I would focus on **procrastination, difficulty keeping track of assignments, and poor time management**. These are common challenges for college students, and each can be addressed with a simple feature that is realistic to implement for a computer science final project.

3 Common Problems and Their Solutions

#	Common Problem	Why It Is a Problem	Simple Website Feature
1	**Procrastination**	Students put off assignments until the last minute, leading to stress and rushed work.	**Task Breakdown:** Allow students to split large assignments into smaller tasks, such as researching, outlining, writing, and proofreading.
2	**Forgetting Assignments and Deadlines**	Students have multiple classes with different assignments and due dates, making it easy to forget important work.	**Assignment Dashboard:** Create a centralized list of assignments organized by class, due date, and completion status.
3	**Poor Time Management**	Students struggle to balance studying, homework, exams, extracurricular activities, and personal responsibilities.	**Daily Study Planner:** Allow students to schedule study sessions and assignments into specific time slots throughout the day.

How Each Feature Would Work

1. Task Breakdown

- Students enter a large assignment, such as a research paper.
- They divide it into smaller tasks with individual deadlines.
- They check off each task as they complete it.

- The website displays their progress toward finishing the assignment.

Example: Instead of seeing "Chemistry Lab Report" as one overwhelming task, students can see four smaller tasks: analyze data, complete calculations, write the report, and proofread.

2. Assignment Dashboard

- Students add assignments for each of their classes.
- Each assignment includes a name, class, due date, and priority level.
- Assignments are automatically sorted by their upcoming deadlines.
- Students can mark assignments as completed.

Example: A student logs in and immediately sees that their chemistry homework is due tomorrow, their English essay is due Friday, and their computer science project is due next week.

3. Daily Study Planner

- Students select assignments they want to work on.
- They assign each task a planned start time and duration.
- The website displays a daily schedule.
- Students can check off completed study sessions.

Example: A student plans to study chemistry from 2:00–3:00 PM, work on an English essay from 3:30–4:30 PM, and review computer science from 7:00–8:00 PM.

Which Features Should You Prioritize?

For a computer science final project, I recommend implementing these features in the following order:

1. **Assignment Dashboard:** The foundation of your website. You can practice storing, displaying, sorting, and updating data.
2. **Task Breakdown:** Adds useful functionality while helping students manage larger assignments.
3. **Daily Study Planner:** Makes the website more personalized and helps students organize their time.

An Additional Idea: Make the Website Stand Out

You could add a simple **progress tracker** that shows the percentage of assignments completed for each class. For example, if a student has completed 8 out of 10 assignments,

the website displays 80% completion.

This would give students a quick overview of their progress without requiring complicated functionality.

Overall goal: Your website should help college students answer three questions whenever they open it:

- **What do I need to finish?** (Assignment Dashboard)
- **What steps do I need to take?** (Task Breakdown)
- **When should I work on it?** (Daily Study Planner)

These three features work together to make the website practical, easy to use, and achievable as a computer science project.

QUESTION: Can we simplify this project so it is easier for a first-year computer science student to make using basic Python concepts and drafter?

*AI: Yes! Since you're a first-year computer science student using **basic Python and Drafter**, I recommend simplifying your website into a basic assignment tracker. You can still solve real problems for college students without needing complicated features like automatic scheduling, databases, or AI.*

The goal is to make something useful while practicing concepts you're learning in computer science.

1. Simplified Project: College Assignment Tracker

***Project idea:** A simple website that allows college students to add assignments, view upcoming deadlines, and mark assignments as completed.*

Instead of trying to build an all-in-one productivity app, focus on solving one central problem: ***helping students keep track of their assignments so they don't forget deadlines.***

2. Three Common Problems and Simple Solutions

Problem	***Simple Solution***	***How It Works***

1. Students forget assignments.	***Assignment List***	*Students enter an assignment name, class, and due date. The website displays all their assignments in a list.*
2. Students struggle to decide what to work on first.	***Upcoming Assignments***	*The website separates assignments that are still pending from those that have been completed. Students can see what work remains.*
3. Students lose track of completed work.	***Completion Button***	*Students click a button to mark an assignment as completed. The website updates its status.*

These features are simple, but together they create a functional study planner.

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3. What Would the Website Look Like?

You could create a simple interface with three sections.

Section 1: Add an Assignment

Students enter information into a form.

For example:

- ***Assignment name:*** *Chemistry Homework 3*
- ***Class:*** *Chemistry 111*
- ***Due date:*** *November 3*
- ***Button:*** *Add Assignment*

When the student clicks the button, the assignment appears in their assignment list.

Section 2: My Assignments

The website displays assignments like this:

My Assignments

Assignment	*Class*	*Due Date*	*Status*
Chemistry Homework	*Chemistry*	*Nov. 3*	*Pending*

3	*111*		
English Essay	*English 110*	*Nov. 5*	*Pending*
Python Lab	*CISC 101*	*Nov. 7*	*Completed*

You don't need an elaborate calendar. A simple list is enough.

Section 3: Mark as Completed

Each pending assignment has a button labeled ***Mark Complete****.*

When the student clicks it, the assignment's status changes from `Pending` *to* `Completed`*.*

You could also add a filter that lets students view only pending assignments or only completed assignments. This is optional.

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4. How to Build It Using Basic Python and Drafter

You can implement the project using concepts commonly taught in an introductory computer science course.

Website Feature	***Python Concept***	***How You Would Use It***
Store assignments	*Lists*	*Keep track of multiple assignments.*
Store assignment details	*Dictionaries or dataclasses*	*Represent an assignment's name, class, due date, and status.*
Add assignments	*Functions*	*Create a function that adds a new assignment.*
Display assignments	*For loops*	*Loop through the assignment list and display each one.*
Mark assignments complete	*Conditionals and list updates*	*Change an assignment's status when a button is clicked.*
Navigate the website	*Drafter pages and buttons*	*Create pages and let users navigate between them.*

Important: *You can start by storing assignments in a Python list. You don't need a database for the first version. Depending on how your Drafter environment handles state, you can store the list in the application's state and update it when users interact with the website.*

5. Keep the Drafter Structure Simple

You could build the website with just two pages.

Page 1: Home Page

- *Website title: College Assignment Tracker*
- *Short welcome message*
- *Button to view assignments*
- *Button to add an assignment*

Page 2: Assignment Page

- *Form for entering assignment details*
- *List of saved assignments*
- *Button to mark assignments complete*
- *Button to return home*

You could also put everything on one page if that is easier with the Drafter concepts you've learned.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.